.INCH-POUND
MIL-PRF-49450/16(CR)
10 March 2000

PERFORMANCE SPECIFICATION SHEET

HARNESS, USED IN BB-664/A RECHARGEABLE, NICKEL-CADMIUM, VENTED, AIRCRAFT BATTERY.

This specification is approved for use by the USACECOM, Department of Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the performance specification, MIL-PRF-49450(CR)

PERFORMANCE.

Operation The Harness assembly shall provide over-temperature warning and have internal heating elements. The heater assembly shall deliver heat uniformly to the battery cells. When tested in a battery, it shall be capable of raising the cell temperature from -65^0 F to 14^0 F within one hour after start of heating.

Interchangeability. The harness assembly shall be 100% interchangeable with the configuration of the BB-664/A Battery.

Operating voltage. The harness shall operate from an input of 115 VAC, 400 Hz,

Connector J2. The harness shall use a connector that meets the dimensional and interface requirements of MS27468T11B35P.

AMSC N/A: FSC 6140

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DRAWING NOTES

All dimensions are in inches, unless specified otherwise.

FOR FIGs. 1& 2, WIRING HARNESS, WIRING DIAGRAM			
REF LTR	COLOR	TERMINATIONS	MIN LENGTH
A	Red	Pin 4, Power + Lug	3.5
B	White	Pin 6, Thermistor	As Required
C	White	Pin 7, Thermistor	As Required
D	White	Pin 2, Cell #10	13
E	Black	Pin 3, Power – Lug	6
F	Green	S1, Power – Lug	12
G	Green	Pin 5, S1	12
H	Yellow	Pin 10, S3	9.5
I	Orange	Pin 12, S3	9
J	NO	LONGER USED	
K	Orange	S2, S3	12
L	Orange	Pin 11, S2	12
M	Yellow	Pin 9, Heater	16.5
N	Yellow	S2, S3	12
O	White	S2, Heater	As Required

Thermoswitches. The harness shall contain three thermoswitches. Thermoswitch contacts shall be isolated from any battery voltage.

Battery Overheat Switch. A thermoswitch shall open when the battery temperature exceeds 57 ± 3°C (135 ± 5° F) and shall reclose when the battery temperature falls below 49 ± 3°C (120° ± 5°F).

Heater Control Thermoswitches. Two (2) series connected single pole double throw SPDT thermoswitches shall be used to control the operation of the battery heater. One thermoswitch mounted on a cell link shall transfer its contacts when the battery temperature decreases to -1.7 ± 3.3°C (29 ± 6°F). Contacts will be transferred back to deactivate the heater when battery temperature rises above 6.7 ± 4.5°C (44 ± 8° F). The temperature differential between the opening and closing point shall be a minimum 5°C. The second thermoswitch mounted on the center heater blanket shall transfer from the hot to cold position when the heater temperature rises to a value not exceeding 110°C (230° F). It shall transfer back to the cold position when the heater temperature falls to a value of 100 ± 5° C (212 ± 8°F).

Cell Voltage Tap. The harness shall provide a connection to the positive terminal of the tenth cell referenced to the battery negative. The voltage measured from this tap to the battery negative terminal shall be 52.6 ± 1.5% of the overall battery voltage during open circuit charged conditions.

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Dielectric Strength. The harness when tested shall show no evidence of flashover, spark over or breakdown. Current shall not exceed 2 milliamperes.

Heaters. The battery harness shall contain a heater assembly. It shall have a total power of 342 ± 15 watts of heat uniformly delivered to the battery cells. When tested in a battery, while operating from an input of 115 VAC, 400 Hz, it shall be capable of raising the cell temperature from -65° F (-54° C) to 14° F (-10° C) within one hour after the start of heating. Heater hot spot temperature shall not exceed 230° F (110° F). When tested the heater shall demonstrate operation for 500 hours without deterioration or breakdown under visual examination or Dielectric (Hi-pot) testing or change of resistance.

Insulation Resistance of harness. The insulation resistance of the harness shall be measured between the metal area of the sensor receptacle (exclude applicable pins) and any current carrying part for one minute.

Heater Operation. The battery shall be charged with constant current at 13.0 amperes to 29.45 ± 0.2 volts, then 2.6 amperes for another 3 hours and stored at -65° F (-54° C) for 16 hours. With the environmental chamber maintained at -65° F, heater power shall be applied for a period of 100 hours. Continuous temperature measurements shall be taken of all temperature switches and transducers as well as cell electrolyte temperature. Current flow measurements shall be taken of the heater power. The cell electrolyte temperature shall exceed 14° F (-10° C) within one hour after the start of heating.

Harness temperature shock. The harness assembly shall be subject to the Thermal Shock Test specified in MIL-PFR-49450.

Low Temperature Operation. The battery shall be charged constant current at 13.0 amperes to 29.45 ± 0.2 volts, then 2.6 amperes for another 3 hours, and then discharged at 6.5 amperes for 1 hour. It shall then be stored at -20° F (-29° C) for 16 hours. Thermocouples shall be placed on all the thermal switches, transducers and in minimum of one cell. While still at -20°F, heater and charging power shall be applied for two hours. Charging input shall be 11.5 ± 0.5 amp until a temperature compensated voltage of 29.5 ± .1 (at 22° C) increased by 50 millivolts for each degree centigrade below 22° C is reached. At this temperature compensated voltage, the charging current shall be decreased to 5.0 ± 0.2 amperes for the remainder of the 2 hour period. The battery shall then be immediately discharged at 40 amperes until an 18.0 volt endpoint is reached. Minimum time to 18.0 volt is 12 minutes.

Step 1. Check resistance between pins 9 and 10 of wiring harness connector. Resistance should be greater than 1 megohm at temperatures above 32° F (0° C).

Step 2. Check resistance between pins 9 and 11 of wiring harness connector. Resistance should be greater than 1 megohm at temperatures above 32° F (0° C).

Step 3. Check resistance between pins 9 and 12 of wiring harness connector. Resistance

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should be between 25 and 50 ohms.

Step 4. Check for continuity between pins 10 and 11 of wiring harness connector. Resistance should be 0.1 ohm maximum at temperatures above 32° F (0° C).

Step 5. Check resistance between pins 11 and 12 of wiring harness connector. Resistance should be greater than 1 megohm at temperature above 32° F (0° C).

Step 6. Check resistance of thermistor, pins 6 and 7 of wiring harness connector. Resistance versus battery temperature should be as specified below.

Temperature	Acceptable Resistance Range (ohms)
50° F (10° C)	2760-2860
60° F (16° C)	2840-2930
70° F (21° C)	2910-2990
80° F (27° C)	2990-3050
90° F (32° C)	3060-3125
100° F (38° C)	3130-3200
110° F (43° C)	3200-3280

Step 7. Check resistance between pins 3 and 5 of wiring harness connector. Resistance should be greater than 1 megohm at battery temperature below 135 ± 9°F (57 ± 9° C).

Step 8. Check battery voltage between pins 3 (negative) and 4 (positive) of wiring harness connector. Voltage should read total battery voltage. Check that red wire and black/green wires are properly connected.

Step 9. Check battery voltage between pins 3 (negative) and 2 (positive) of wiring harness connector. Voltage should read half the battery voltage obtained in step 8 above. Check that white wire is properly connected to the 10th cell from the negative end of battery.

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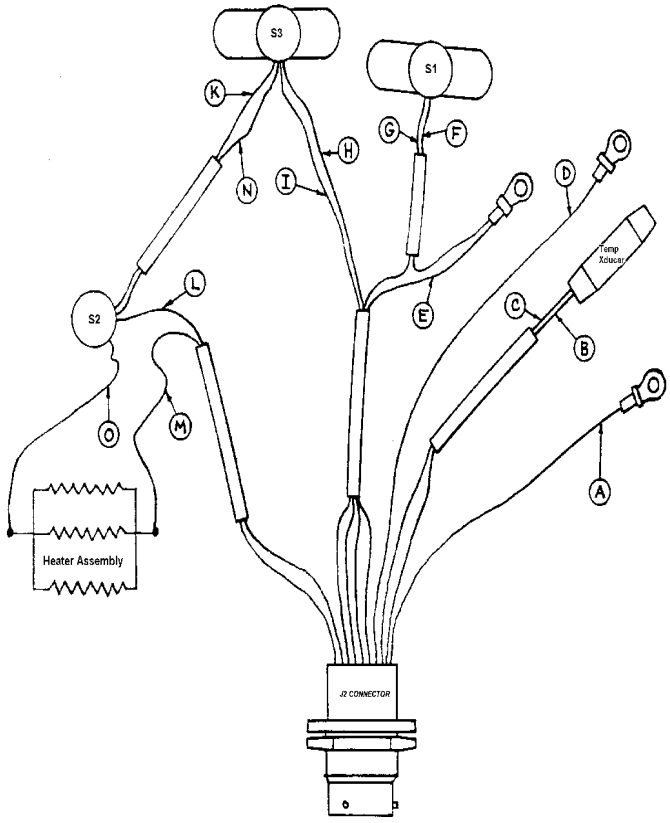


Figure 1. Battery BB-664/A Typical Wiring Harness Configuration

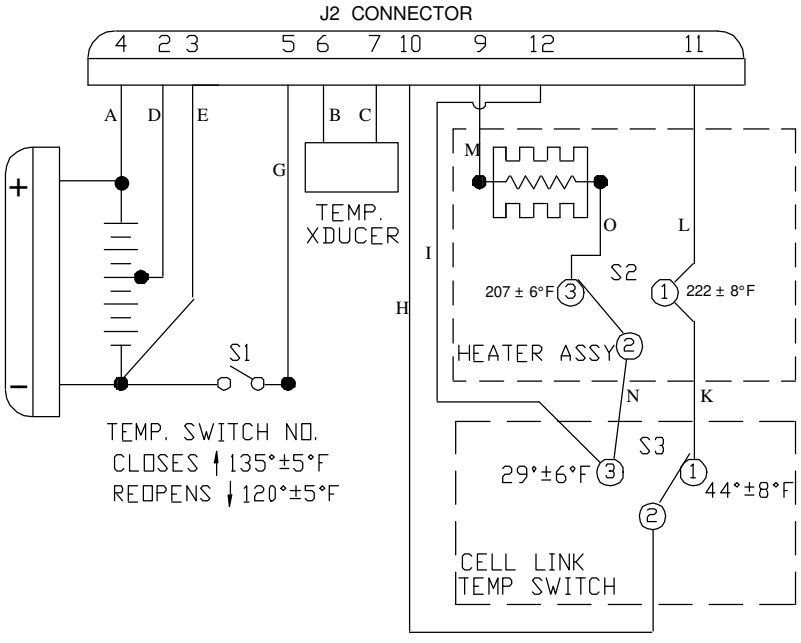


Figure 2. Battery BB-664/A Wiring Harness, Wiring Diagram

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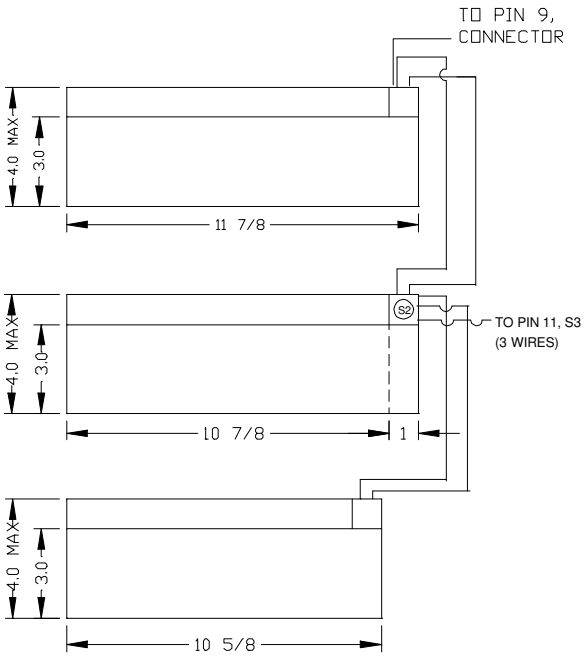


Figure 3. Battery BB-664/A Wiring Harness Heater Assembly

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